Drugs & Aging (2022) 39:305–311
https://doi.org/10.1007/s40266-022-00932-3

ORIGINAL RESEARCH ARTICLE

# Postoperative Pain Treatment in Patients with Dementia: A Retrospective Observational Study

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Accepted: 1 March 2022 / Published online: 1 April 2022
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## Abstract

**Background** Inadequate postoperative analgesia has been noted in patients with dementia, but this topic has only been studied in hip fractures. This study aimed to examine whether the duration of postoperative analgesia associated with three surgical procedures was shorter in patients with dementia than in those without dementia.

**Methods** This retrospective observational study was based on a nationwide discharge database of acute care hospitals in Japan and included patients aged ≥ 65 years diagnosed with lung cancer, rectal cancer, or hip fracture, who underwent surgery in 366 hospitals between April 2013 and September 2018. The primary outcome was the incidence of injectable analgesic use during the postoperative hospital stay. The number of person-days of opioid, non-steroidal anti-inflammatory drug, or acetaminophen use was calculated for patients with and without dementia after each surgery using generalized estimating equations to obtain the age-adjusted incidence per 100 person-days.

**Results** Among the 32,379 patients included, 4828 (14.9%) had dementia. The duration of opioid administration per 100 person-days was 14% shorter in patients with dementia than in those without dementia after open rectal cancer surgery (incidence rate ratio [IRR] 0.86; 95% confidence interval [CI] 0.74–1.00) and 20% shorter in patients with dementia after open lung cancer surgery (IRR 0.80; 95% CI 0.70–0.91). In patients who underwent thoracoscopic lung cancer surgery, laparoscopic rectal cancer surgery, or hip replacement surgery, the duration of opioid administration was shorter in patients with dementia than in those without dementia, but the difference was not statistically significant.

**Conclusions** The length of postoperative opioid administration after lung and rectal cancer surgery was reduced in patients with dementia, suggesting that the duration of postoperative analgesia in patients with dementia may be insufficient. More careful pain assessment and management of postoperative analgesia in these patients may be necessary.

## 1 Introduction

In today's aging global population, the number of patients with dementia is increasing [1].

Patients with dementia experience pain, although pain thresholds and methods of expression may vary between patients with and without dementia. Therefore, the demand for pain management in patients with dementia is increasing; however, pain assessment in these patients remains challenging. One reported reason for this challenge is that people with dementia are unable to effectively communicate pain, making it difficult for medical professionals to detect pain in these patients [2]. In addition, with regard to the use of analgesics, especially opioids, concerns about enhanced side effects in patients with dementia have caused physicians to withhold their use [3]. Therefore, previous studies have suggested that pain management could be insufficient in patients with dementia [3–6].

Postoperative pain management is particularly difficult in patients with dementia owing to their declining cognitive function and the behavioral and psychological symptoms of dementia [5]. Providing adequate pain control is important because residual postoperative pain can cause delirium, decrease respiratory function, and prolong the hospital stay [7, 8]. However, previous findings have shown that after

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**Key Points**

The duration of postoperative analgesia was shorter in patients with dementia than in those without dementia.

This difference may be more apparent in painful surgeries, such as an open surgery, than in endoscopic surgery.

Healthcare professionals should be aware of difficulties in assessing postoperative pain in patients with dementia and be especially attentive to its management, as well as to the prevention of pain-related adverse events.

hip fracture surgery, patients with dementia received lower doses of opioids than those without dementia [9]. This suggests that postoperative analgesia in patients with dementia may be inadequate [10].

To date, only studies on hip and pelvic fractures have examined a potential association between dementia and postoperative analgesic use [9]. To the best of our knowledge, this association has not been studied in other surgeries. Moreover, most studies only analyzed postoperative analgesia using doses expressed in morphine equivalents. However, previous studies have shown that patients with dementia are generally likely to have poorer renal function and lower body weight and albumin levels owing to the presence of sarcopenia than patients without dementia. Therefore, evaluation of the dosage of morphine, which varies according to renal function and body weight, is not an adequate assessment [9].

Therefore, in this study, we compared the duration of analgesic use after multiple surgeries in patients with and without dementia to determine whether the extent of postoperative analgesia was similar in both groups.

## 2 Methods

### 2.1 Data Source

We conducted a retrospective cohort study using a nationwide discharge abstract database maintained by Medical Data Vision Co., Ltd. In the study period, between April 2013 and March 2018, 366 acute care hospitals in Japan were included in the database. Details of this database have been previously described [11, 12]. Briefly, the database contains clinical and procedural information of all inpatients from the participating hospitals. All patients were de-identified and guaranteed anonymity. Owing to the anonymous nature of the data, the requirement for institutional review board approval was waived.

### 2.2 Study Population

We identified patients aged ≥ 65 years admitted to acute care hospitals from April 1, 2013 until March 31, 2018. We selected patients whose most 'resource-consuming diagnoses' during hospitalization were lung cancer, rectal cancer, or hip fracture, and who had undergone surgical therapy via laparotomy/thoracotomy, laparoscopy/thoracoscopy, or hip replacement surgery, respectively.

Patients were divided into five groups based on the diagnosis and type of surgery: open surgery for colorectal cancer, laparoscopic surgery for colorectal cancer, open surgery for lung cancer, thoracoscopic surgery for lung cancer, and hip replacement surgery for hip fractures. We defined the index date as the day on which the patient underwent surgery at an acute care hospital.

### 2.3 Study Outcome

The primary outcome was the number of days of injectable analgesic use in the period from the index date to the discharge date. The analgesics used in the postoperative period were classified into three groups as follows: opioids, non-steroidal anti-inflammatory drugs (NSAIDs), and acetaminophen. For each drug, we distinguished the mode of administration. For injectable opioids and NSAIDs/acetaminophen, we calculated the number of person-days during which the drug was used in each surgical group. We focused only on injectable drugs because injectable opioids are the main analgesic agents used after the surgeries considered in this study and injectable NSAIDs/acetaminophen are used as substitutes for injectable opioids. Person-days were calculated instead of doses because while the dose depends on an individual's weight and renal function, person-days are less affected by these factors, allowing for a more accurate comparison among individuals.

### 2.4 Primary Explanatory Variable and Covariate

The primary explanatory variable assessed was the dementia status. Prevalent dementia was defined as at least one prescription of an anti-dementia drug during hospitalization (i.e., donepezil, galantamine, memantine, or rivastigmine), a diagnosis of dementia (International Statistical Classification of Diseases and Related Health Problems 10th Revision [ICD-10] codes: F00–F03, F05.1, and G30–G31), or a score of I or greater on the Dementia Scale [13], an observer-rated measure used to assess the degree of independence in activities of daily living (ADL) owing to dementia. The Dementia Scale has a high degree of sensitivity and specificity for identifying dementia via a

Clinical Dementia Rating [14], good predictive validity for changes in ADL assessed by the Functional Assessment Staging Test [15], and sufficient criterion validity against the Mini-Mental State Examination and the Hasegawa Dementia Scale–Revised [16]. All hospitals contributing to the DPC database were asked by the attending physician to rate patients using the Dementia Scale at the timing of hospital admission, which is recorded and excludes transient disturbances owing to physical conditions. Age was used as a covariate in the analysis, categorized in 5-year increments as follows: 65–69, 70–74, 75–79, 80–84, and 85+ years.

### 2.5 Statistical Analysis

Descriptive statistics are reported as median and interquartile range. The number of person-days of opioid, NSAID, or acetaminophen administration was calculated for patients with and without dementia in each surgery category. The age-adjusted incidence rate in patients with versus without dementia was compared using Poisson regression with generalized estimating equations (GEEs). Incidence rate ratios (IRRs) are presented with their 95% confidence interval (CI). All statistical analyses were performed using R version 4.0.3 (R Foundation for Statistical Computing, Vienna, Austria). GEE models were fitted using the geeglm function in the geepack R package [17]. A significance level of 5% was used in all analyses.

## 3 Results

A total of 32,379 patients included in the dataset were eligible for analysis, among whom 4828 (14.9%) had dementia (Table 1). Except for hip replacement surgery, in all surgeries examined, including lung cancer surgery (open and thoracoscopic) and colorectal cancer surgery (open and laparoscopic), the percentage of opioid prescription for postoperative pain in the study population for each surgery was higher than 90% (Table 2). In hip replacement surgery, opioids were used by 62.9% of individuals without dementia and 57.7% of those with dementia (Table 2). Among the opioids administered, fentanyl injection was most commonly used (70% of all opioids administered). Among all opioid drugs, 91.7% were administered via injection (Supplementary Table 1, see the electronic supplementary material).

The duration of opioid administration per 100 person-days in patients with dementia was 14% shorter than in those without dementia in open surgery for rectal cancer (IRR 0.86; 95% CI 0.74–1.00) and 20% shorter in open surgery for lung cancer (IRR 0.80; 95% CI 0.70–0.91; Table 3). In both thoracoscopic surgery for lung cancer and laparoscopic surgery for rectal cancer, the duration of opioid administration was shorter in patients with dementia (IRR 0.94; 95% CI 0.87–1.02 and IRR 0.93; 95% CI 0.84–1.04, respectively), but the difference was not statistically significant. Similarly, an assessment of postoperative treatment for hip replacement

**Table 1** Characteristics of patients who underwent surgical procedures

| Characteristics | Open rectal cancer surgery (*n* = 2115) | Rectal cancer laparoscopic surgery (*n* = 5148) | Open lung cancer surgery (*n* = 2481) | Lung cancer thoracoscopic surgery (*n* = 14,784) | Hip replacement surgery (*n* = 7851) |
|---|---|---|---|---|---|
| Sex, *n* (%) | | | | | |
| Male | 1272 (60.1) | 3212 (62.4) | 1698 (68.4) | 8703 (58.9) | 1737 (22.1) |
| Female | 843 (39.9) | 1936 (37.6) | 783 (31.6) | 6081 (41.1) | 6114 (77.9) |
| Age, *n* (%) | | | | | |
| 65–69 | 670 (31.7) | 1724 (33.5) | 794 (32.0) | 4347 (29.4) | 530 (6.8) |
| 70–74 | 520 (24.6) | 1345 (26.1) | 768 (31.0) | 4338 (29.3) | 768 (9.8) |
| 75–79 | 453 (21.4) | 1100 (21.4) | 594 (23.9) | 3567 (24.1) | 1258 (16.0) |
| 80–84 | 296 (14.0) | 669 (13.0) | 267 (10.8) | 2031 (13.7) | 1896 (24.1) |
| ≥ 85 | 176 (8.3) | 310 (6.0) | 58 (2.3) | 501 (3.4) | 3399 (43.3) |
| Dementia, *n* (%) | | | | | |
| No | 1984 (93.8) | 4885 (94.9) | 2322 (93.6) | 14,269 (96.5) | 4091 (52.1) |
| Yes | 131 (6.2) | 263 (5.1) | 159 (6.4) | 515 (3.5) | 3760 (47.9) |
| Length of stay after surgery, days, median (IQR) | 15 (12–18) | 12 (10–15) | 10 (8–13) | 8 (6–11) | 16 (14–20) |

*IQR* interquartile range

**Table 2** Prevalence of injectable analgesic use stratified by dementia status

| Surgery and dementia | *N* | Administration, % | | |
|---|---|---|---|---|
| | | Opioid | NSAIDs | Acetaminophen |
| Open rectal cancer surgery | | | | |
| Without dementia | 1984 | 97.5 | 77.6 | 43.1 |
| With dementia | 131 | 98.5 | 70.2 | 44.3 |
| Rectal cancer laparoscopic surgery | | | | |
| Without dementia | 4885 | 97.6 | 74.1 | 49.4 |
| With dementia | 263 | 96.6 | 72.2 | 50.2 |
| Open lung cancer surgery | | | | |
| Without dementia | 2322 | 98.8 | 94.3 | 45.1 |
| With dementia | 159 | 99.4 | 98.7 | 23.9 |
| Lung cancer thoracoscopic surgery | | | | |
| Without dementia | 14269 | 97.8 | 90.6 | 45.1 |
| With dementia | 515 | 98.1 | 88.9 | 35.7 |
| Hip replacement surgery | | | | |
| Without dementia | 4091 | 62.9 | 77.2 | 38.7 |
| With dementia | 3760 | 57.7 | 70.2 | 35.2 |

Counted if used at least once during hospitalization

*NSAID* non-steroidal anti-inflammatory drug

surgery revealed that the group with dementia received opioids for a shorter duration than the group without dementia, but the difference was not statistically significant (IRR 0.96; 95% CI 0.91–1.02).

Patients with dementia were 18% more likely to receive NSAIDs/acetaminophen injections after open surgery for lung cancer than those without dementia (IRR 1.18; 95% CI 1.03–1.35; Table 3). For all surgeries, except open surgery for lung cancer, the duration of NSAIDs/acetaminophen administration was shorter in patients with dementia than in those without dementia. However, the difference was statistically significant only for patients who underwent hip replacement surgery.

## 4 Discussion

This study examined the differences in analgesic use for postoperative pain management between patients with and without dementia. The results showed that the duration of postoperative analgesia was likely to be shorter in patients with dementia who underwent open surgery for lung and rectal cancer. Furthermore, the duration of NSAIDs/acetaminophen, used for replacement of opioids, was found to be shorter or equivalent in patients with dementia than those without dementia for the surgeries other than open surgery for lung cancer. These results suggest that patients with dementia may receive less postoperative analgesia after some surgeries than those who are unaffected by the disease.

Previous research reported that in patients with hip fractures, levels of morphine-equivalent postoperative analgesia were lower in patients with dementia than those without dementia [9, 18]. In this study, after hip replacement surgery, the number of days of opioid administration was shortened for patients with dementia, although not at a statistically significant level. However, the duration of NSAIDs/acetaminophen administration was significantly shorter in patients with dementia than in those without dementia. Altogether, the findings suggest that postoperative analgesia for hip replacement surgery is less extensively used in patients with dementia, which is consistent with findings from previous studies.

The present study showed that the reduction in the duration of opioid administration after thoracoscopic/laparoscopic surgery in patients with dementia was less pronounced than that after open thoracic/abdominal surgery. This is probably because postoperative pain after thoracoscopic or laparoscopic surgery is lower than that after open surgery [19, 20], making opioid use less necessary. This may be the reason why no statistically significant differences in the duration of opioid administration were observed between patients with and without dementia in thoracoscopic or laparoscopic surgeries. On the other hand, painful surgeries, such as open surgery, were significantly associated with differences in the duration of opioid administration between patients with and without dementia.

Previous studies have suggested that the reduced duration of opioid use is mainly due to the inability of patients

**Table 3** Incidence rate ratio of injectable analgesic use in patients with and without dementia

| Surgery and dementia | Person-days | Opioids | | | | NSAIDs/acetaminophen | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Prescription days | Crude incidence per 100 person-days | Adjusted incidence per 100 person-days | Incidence rate ratio (95% CI) | Prescription days | Crude incidence per 100 person-days | Adjusted incidence per 100 person-days | Incidence rate ratio (95% CI) |
| Open rectal cancer surgery | | | | | | | | | |
| Without dementia | 30,368 | 3526 | 11.6 | 11.2 | 0.86 (0.74–1.00)* | 4133 | 13.6 | 13.2 | 0.95 (0.83–1.08) |
| With dementia | 2054 | 193 | 9.4 | 9.6 | | 253 | 12.3 | 12.6 | |
| Rectal cancer laparoscopic surgery | | | | | | | | | |
| Without dementia | 64,161 | 7174 | 11.2 | 10.9 | 0.93 (0.84–1.04) | 8284 | 12.9 | 12.7 | 0.99 (0.90–1.09) |
| With dementia | 3675 | 369 | 10.0 | 10.2 | | 457 | 12.4 | 12.6 | |
| Open lung cancer surgery | | | | | | | | | |
| Without dementia | 25,572 | 3873 | 15.1 | 14.2 | 0.80 (0.70–0.91)* | 2653 | 10.4 | 10.2 | 1.18 (1.03–1.35)* |
| With dementia | 1896 | 226 | 11.9 | 11.3 | | 231 | 12.2 | 12.0 | |
| Lung cancer thoracoscopic surgery | | | | | | | | | |
| Without dementia | 129,499 | 18,420 | 14.2 | 13.9 | 0.94 (0.87–1.02) | 12,909 | 10.0 | 9.8 | 0.91 (0.83–1.00) |
| With dementia | 4986 | 654 | 13.1 | 13.1 | | 441 | 8.8 | 8.9 | |
| Hip replacement surgery | | | | | | | | | |
| Without dementia | 69,589 | 2807 | 4.0 | 4.1 | 0.96 (0.91–1.02) | 2377 | 3.4 | 3.5 | 0.90 (0.85–0.96)* |
| With dementia | 62,545 | 2333 | 3.7 | 4.0 | | 1874 | 3.0 | 3.2 | |

*CI* confidence interval, *NSAID* non-steroidal anti-inflammatory drug

*Indicates statistical significance

with dementia to effectively communicate their pain and the inability of medical professionals to accurately assess pain in these patients, leading physicians to inaccurately conclude that opioids are unnecessary [2, 4]. However, another reason could be that physicians refrain from prescribing opioids to patients with dementia owing to their potential to cause delirium and disorientation [21]. In this regard, NSAIDs/acetaminophen are valid alternatives to opioids. In fact, for open chest lung cancer surgery, the duration of NSAIDs/acetaminophen administration was prolonged in patients with dementia compared with that in patients without dementia. However, after hip replacement surgery, the duration of NSAIDs/acetaminophen administration in patients with dementia was shortened relative to patients without dementia. The observed differences may be related, as mentioned above, to differences in the levels of pain associated with each surgical technique. These findings require

validation in further studies using actual pain scale measurements of patients and physician awareness surveys.

The use of analgesics in postoperative analgesia may be related to the severity and subtype of dementia. In fact, there is some discussion that the pain threshold may be raised in Alzheimer's disease [22, 23]. However, in the present study, the limitations of the data and the small difference in the incidence of postoperative analgesic use between patients with and without dementia make it difficult to evaluate these issues. Furthermore, among the three health issues studied (hip fracture, rectal cancer, and lung cancer), patients with cancer have different patient characteristics, such as age, from those with hip fracture. In addition, most surgeries for hip fracture are unplanned, whereas cancer surgeries are planned. Therefore, the results of the hip fracture group and the results of the cancer surgery groups should be noted as results from different populations and contexts. In particular, the degree of dementia may differ between patients with dementia in the hip fracture group and those with cancer who choose surgical treatment, and the results in the cancer surgery group cannot be directly compared with those in the hip fracture group. It is hoped that future studies will examine these differences using appropriate pain assessment tools, taking into account the severity of dementia.

Since pain in patients with dementia is difficult to assess, medical professionals should be particularly aware and efforts should be made to accurately estimate the pain level. In this regard, previous studies have reported that developing a scale based on nonverbal complaints, devising appropriate questioning methods, and performing frequent questioning are important [24]. In the future, these interventions should be practiced, particularly when treating patients with dementia who have undergone painful surgeries, and increased emphasis should be placed on devising pain management strategies for patients with dementia.

This study has several limitations. First, since the database used did not accurately report the duration of administration of oral medications, the study was limited to injectable opioids and NSAIDs/acetaminophen. Since oral NSAIDs are used as an alternative to opioid, NSAID, and acetaminophen injection, the duration of postoperative analgesia may have been underestimated. Furthermore, even if the postoperative analgesic period was short, we cannot necessarily assert that postoperative analgesia was inadequate because the patients may have switched to oral medications earlier, which we have not analyzed. Second, since this was a database study, there was no information on whether patients complained of pain or on the patients' general condition, including laboratory data, resulting in a lack of information regarding factors responsible for the reduced duration of analgesic administration. Further research using information from medical records is needed. Third, although the database used includes data from 366 acute care hospitals that agreed to provide them, it does not include all acute care hospitals in Japan. Therefore, the degree to which results are representative of the Japanese population is limited.

In conclusion, in some surgeries, the length of postoperative opioid administration was shorter in patients with dementia, suggesting that the duration of postoperative analgesia in patients with dementia may be shorter than that in patients without dementia. More careful pain assessment and management of postoperative analgesia are needed in the treatment of patients with dementia, especially after painful surgeries.

**Supplementary Information** The online version contains supplementary material available at https://doi.org/10.1007/s40266-022-00932-3.

## Declarations

**Funding** This study was supported by the Japan Agency for Medical Research and Development (No: 18dk0207024h0003). The sponsors of this study had no role in the design and conduct of the study; collection, management, analysis, or interpretation of the data; preparation, review, or approval of the manuscript; or the decision to submit the manuscript for publication.

**Conflict of interest** The authors have no conflicts of interests.

**Ethics approval** Not applicable.

**Data availability** The data are not publicly available because they contain information that could compromise research participant privacy/consent.

**Consent to participate** Not applicable.

**Consent for publication** Not applicable.

**Code availability** The R codes used for the study are available from the corresponding author on request.

**Author contributions** NS, YO, and AO: Study concept and design. YO: Analysis of the data. NS, YO, AO: Interpretation of the data. NS and YO: Preparation of the manuscript. AO: Critical revision of the manuscript.

## References

1. OECD. Health at a glance 2019. Paris: OECD iLibrary; 2019. p. 243.
2. Smigorski K, Leszek J. Pain experience and expression in patients with dementia, in health management. London, United Kingdom: IntechOpen; 2010 [Online]. https://www.intechopen.com/chapters/11528. https://doi.org/10.5772/9886.
3. Rantala M, Kankkunen P, Kvist T, Hartikainen S. Barriers to postoperative pain management in hip fracture patients with dementia as evaluated by nursing staff. Pain Manag Nurs. 2014;15:208–19. https://doi.org/10.1016/j.pmn.2012.08.007.
4. Rantala M, Hartikainen S, Kvist T, Kankkunen P. Analgesics in postoperative care in hip fracture patients with

dementia—reported by nurses. J Clin Nurs. 2014;23:3095–106. https://doi.org/10.1111/jocn.12548.
5. Rantala M, Kankkunen P, Kvist T, Hartikainen S. Post-operative pain management practices in patients with dementia—the current situation in Finland. Open Nurs J. 2012;6:71–81. https://doi.org/10.2174/1874434601206010071.
6. Shen C, Zhao X, Dwibedi N, Wiener RC, Findley PA, Sambamoorthi U. Opioid use and the presence of Alzheimer's disease and related dementias among elderly Medicare beneficiaries diagnosed with chronic pain conditions. Alzheimers Dement (NY). 2018;4:661–8. https://doi.org/10.1016/j.trci.2018.10.012.
7. Morrison SR, Magaziner J, McLaughlin MA, Orosz G, Silberzweig SB, Koval KJ, et al. The impact of post-operative pain on outcomes following hip fracture. Pain. 2003;103:303–11. https://doi.org/10.1016/S0304-3959(02)00458-X.
8. Proc JL, Jordan H, Docherty AB. Perioperative care of people with dementia. Br J Hosp Med (Lond). 2020;81:1–9. https://doi.org/10.12968/hmed.2019.0345.
9. Moschinski K, Kuske S, Andrich S, Stephan A, Gnass I, Sirsch E, et al. Drug-based pain management for people with dementia after hip or pelvic fractures: a systematic review. BMC Geriatr. 2017;17:54. https://doi.org/10.1186/s12877-017-0446-z.
10. Jensen-Dahm C, Palm H, Gasse C, Dahl JB, Waldemar G. Postoperative treatment of pain after hip fracture in elderly patients with dementia. Dement Geriatr Cogn Disord. 2016;41:181–91. https://doi.org/10.1159/000444704.
11. Suzuki M, Arai M, Hayashi A, Ogino M. Prescription pattern of anti-Parkinson's disease drugs in Japan based on a nationwide medical claims database. eNeurologicalSci. 2020;20:100257. https://doi.org/10.1016/j.ensci.2020.100257.
12. Mori A, Hashimoto K, Koroki Y, Wu DB, Masumori N. The correlation between metastasis-free survival and overall survival in non-metastatic castration resistant prostate cancer patients from the Medical Data Vision claims database in Japan. Curr Med Res Opin. 2019;35:1745–50. https://doi.org/10.1080/03007995.2019.1619543.
13. Takigawa Y. Criteria for evaluating the degree of independence of dementia elderly performing activities of daily Living. J Public Health Pract. 1994;58:73–5.
14. Meguro K, Tanaka N, Kasai M, Nakamura K, Ishikawa H, Nakatsuka M, et al. Prevalence of dementia and dementing diseases in the old-old population in Japan: the Kurihara Project. Implications for Long-Term Care Insurance data. Psychogeriatrics. 2012;12:226–34. https://doi.org/10.1111/j.1479-8301.2012.00406.x.
15. Arai H, Sumitomo K, Sakata Y, Daidoji K, Takase T, Toyoda T. Disease state changes and safety of long-term donepezil hydrochloride administration in patients with Alzheimer's disease: interim results from the long-term, large-scale J-GOLD study in Japan. Psychogeriatrics. 2016;16:107–15. https://doi.org/10.1111/psyg.12130.
16. Hisano S. The relationship between revised Hasegawa Dementia Scale (HDS-R), Mini-Mental State Examination (MMSE), Bed-fast Scale, and Dementia Scale. Jpn J Geriatr Psychiatry. 2009;20:883–91.
17. Højsgaard S, Halekoh U, Yan J. Generalized estimating equation package; 2016. https://mran.microsoft.com/snapshot/2016-03-13/web/packages/geepack/geepack.pdf Accessed 12 Nov 2021.
18. Feldt KS, Ryden MB, Miles S. Treatment of pain in cognitively impaired compared with cognitively intact older patients with hip-fracture. J Am Geriatr Soc. 1998;46:1079–85. https://doi.org/10.1111/j.1532-5415.1998.tb06644.x.
19. Bates AT, Divino C. Laparoscopic surgery in the elderly: a review of the literature. Aging Dis. 2015;6:149–55. https://doi.org/10.14336/AD.2014.0429.
20. Bendixen M, Jørgensen OD, Kronborg C, Andersen C, Licht PB. Postoperative pain and quality of life after lobectomy via video-assisted thoracoscopic surgery or anterolateral thoracotomy for early stage lung cancer: a randomised controlled trial. Lancet Oncol. 2016;17:836–44. https://doi.org/10.1016/S1470-2045(16)00173-X.
21. Titler MG, Herr K, Schilling ML, Marsh JL, Xie XJ, Ardery G, et al. Acute pain treatment for older adults hospitalized with hip fracture: current nursing practices and perceived barriers. Appl Nurs Res. 2003;16:211–27. https://doi.org/10.1016/s0897-1897(03)00051-x.
22. Binnekade TT, Van Kooten J, Lobbezoo F, Rhebergen D, Van der Wouden JC, Smalbrugge M, et al. Pain experience in dementia subtypes: a systematic review. Curr Alzheimer Res. 2017;14:471–85. https://doi.org/10.2174/1567205013666160602234109.
23. Scherder E, Bouma A, Borkent M, Rahman O. Alzheimer patients report less pain intensity and pain affect than non-demented elderly. Psychiatry. 1999;62:265–72. https://doi.org/10.1080/00332747.1999.11024871.
24. Kelley AS, Siegler EL, Reid MC. Pitfalls and recommendations regarding the management of acute pain among hospitalized patients with dementia. Pain Med. 2008;9:581–6. https://doi.org/10.1111/j.1526-4637.2008.00472.x.